Total No. of Questions : 8]

SEAT No. :

PE-2526

[Total No. of Pages : 2

[6583]-52

T.E. (AI & DS)

ARTIFICIAL NEURAL NETWORK

(2019 Pattern) (Semester - VI)(317531)

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the candidates:

1) ***Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6 and Q.7 or Q.8.***
2) ***Figures to the right side indicate full marks.***
3) ***Neat diagrams must be drawn wherever necessary.***
4) ***Assume sutiable data if necessary.***

Q1) a) What is Hopfield Neural Network? What is a state transition Diagram for Hopfield Neural Network? Explain how to derive it in Hopfield model. **[9]**

b) A hetero associative network is given. Find the weight matrix and test the network with the training input vectors. S1 = (1,1,0,0), S2 = (0,1,0,0), S3 = (0,0,1,1) S4 = (0,0,1,0) t1 = (1,0), t2 = (1,0), t3 = (0,1), t4 = (0,1) **[9]**

OR

Q2) a) Describe Boltzmann Machine and Boltzmann learning Law. What are the limitations of Boltzmann learning. **[9]**

b) For a given input vector (S1,S2,S3,S4) and output vector T=(T1,T2). Find the weight matrix using hetero associative training algorithm.
S1 = (1,1,0,0), S2 = (1,0,0,1), S3 = (1,1,0,0) S4 = (0,1,0,1) t1 = (0,1), t2 = (1,1), t3 = (1,0), t4 = (0,0) **[9]**

Q3) a) Explain ART under the following headings : **[8]**

i) Architecture

ii) Working

iii) Training

iv) Implementation

b) Describe the self-organization map (SOM) algorithm and explain how it can be used for feature mapping. **[9]**

OR

Q4) a) Discuss in detail the following with diagram: **[8]**

i) Learning vector quantization

ii) Adaptive pattern classification

b) Draw the architecture of Kohonen Network and explain the algorithm for training the weights of the Network. **[9]**

Q5) a) Consider a LeNet-5 a convolutional neural network, we want to perform the classification of digits, Write down the complete procedure followed in its architecture. **[9]**

b) Draw and explain the architecture of Convolutional Neural Network. **[5]**

c) Distinguish between CNN and RNN **[4]**

OR

Q6) a) Exemplify convolution over volume with convolution on RGB images. Also illustrate multiple filters used in it. **[9]**

b) Explain the softmax regression with respect to hypothesis and cost function and write down its properties. **[5]**

c) Distinguish between Pooling and Padding. **[4]**

Q7) a) Discuss the following applications of ANN in detail. **[9]**

i) Recognition of consonant vowel (CV) segments.

ii) Recognition of Olympic games symbols.

iii) Recognition of handwritten characters

b) You have been asked to develop a model of recognizing hand written digits. What are the chosen steps for activity? Explain each with detail. **[8]**

OR

Q8) a) Write short note on the following. **[9]**

i) NET Talk

ii) Texture classification

iii) Pattern classification

b) Describe the Neocognitron model and its significance in the recognition of handwritten characters. **[8]**

♣ ♣ ♣